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Improving products of anaerobic sludge digestion by microaeration

P. Jenicek, C. A. Celis, L. Krayzelova, N. Anferova and D. Pokorna

ABSTRACT

Biogas, digested sludge and sludge liquor are the main products of anaerobic sludge digestion. Each of the products is influenced significantly by specific conditions of the digestion process. Therefore, any upgrade of the digestion technology must be considered with regard to quality changes in all products. Microaeration is one of the methods used for the improvement of biogas quality. Recently, microaeration has been proved to be a relatively simple and highly efficient biological method of sulfide removal in the anaerobic digestion of biosolids, but little attention has been paid to comparing the quality of digested sludge and sludge liquor in the anaerobic and microaerobic digestion and that is why this paper primarily deals with this area of research. The results of the long-term monitoring of digested sludge quality and sludge liquor quality in the anaerobic and microaerobic digesters suggest that products of both technologies are comparable. However, there are several parameters in which the 'microaerobic' products have a significantly better quality such as: sulfide (68% lower) and soluble chemical oxygen demand (COD) (33% lower) concentrations in the sludge liquor and the lower foaming potential of the digested sludge.

Key words | anaerobic digestion, biogas desulfurization, microaerobic conditions, microbial activity, sludge liquor, sludge quality

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INTRODUCTION

Currently, anaerobic digestion is a key technological stage of sludge management in large municipal wastewater treatment plants, enabling the transformation of organic pollution into energy. Anaerobic digestion is the only process with a highly positive energy balance and the produced biogas can cover a substantial part of wastewater treatment energy requirements (Jenicek *et al.* 2012). Over the past few decades many intensification methods were proposed for focusing on an increase in biogas production, because the biodegradability of sludge organic matter is limited. In addition to biogas production maximization, another important aim of advanced sludge technologies lies in the quality improvement of all anaerobic digestion products: biogas, sludge and sludge liquor.

Biogas is utilized as a renewable energy source and its sufficient quality determines its energy valorization. In particular, hydrogen sulfide (Lens *et al.* 1998) and siloxane (Dewil *et al.* 2006) content can cause many operational problems. Sludge disposal is usually a costly operation and so the reuse of sludge is strongly encouraged because of sustainability. In terms of sludge reuse routes, the content of specific pollutants, rheological properties, dewaterability and other sludge properties are of great importance.

Sludge liquor separated during digested sludge dewatering is rejected into the activated sludge system and has a considerable influence on wastewater treatment operational costs and the effluent quality. Thus, any improvement to sludge liquor quality can be beneficial for the wastewater treatment process.

One of the ways to improve anaerobic digestion products seems to lie in the application of microaerobic conditions. In this paper, microaerobic digestion means a digestion with limited (trace) oxygen consumption. With respect to the oxidation-reduction potential (ORP), microaerobic conditions can be characterized by a limited ORP increase caused by micro-consumption of oxygen in comparison with anaerobic conditions (Khanal & Huang 2003). The microaerobic conditions are obtained by dosing a limited amount of air or oxygen into the anaerobic digester. It has been proved that in a mixed culture, even strict

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anaerobes can survive without any inhibition, supposing that facultative microorganisms are able to consume the present oxygen quickly and completely (Zitomer & Shrout 1998; Jenicek *et al.* 2011).

Until now microaerobic conditions have been used for efficient hydrogen sulfide removal from biogas. (Khanal & Huang 2003; van der Zee *et al.* 2007; Fdz-Polanco *et al.* 2009). However, less attention has been paid to the comparison of digested sludge and sludge liquor quality after anaerobic and microaerobic digestion. This paper seeks to fill this gap in knowledge by evaluating the quality of all digestion products.

MATERIAL AND METHODS

Digesters set-up

Our experiments were carried out in two continuously stirred, semi-continuously fed laboratory-scale digesters of 10 L working volume, identical to those used in our previous microaerobic experiments reported earlier (Jenicek *et al.* 2010). The total time of parallel operations for both digesters was 300 days. The start-up of both reactors was carried out in anaerobic conditions. After the start-up, the operation of the first reactor (R-1) was changed to microaerobic mode when air was dosed continuously to the bottom of the digester with a peristaltic pump. The air flow rate was gradually increased from 0.3 to 1.6 L/d. The second reactor (R-2) remained anaerobic and served as a referential reactor. Thickened waste activated sludge from a municipal wastewater treatment plant was used as the substrate for the reactors; sulfur content was increased by natrium sulfate addition. The average sludge composition was as follows: total suspended solids (TSS) 69.2 ± 6.5 g/L, volatile suspended solids (VSS) 49.4 ± 3.6 g/L, pH 7.10 ± 0.19. The reaction mixture was kept homogeneous by means of mechanical mixing and the operational temperature was kept at $40 \pm 1\,^{\circ}C$ as it was the original operational temperature of the inoculum. Volumetric loading rates of the digesters were 2.0 and 0.15 g/L.d for chemical oxygen demand (COD) and SO_4^{2-} respectively during the evaluation period. Both of the reactors had equal hydraulic and solids retention times: ca. 30 days.

Biogas and sludge composition

Analytical procedures were carried out in accordance with the *Standard Methods for the Examination of Water and Wastewater* (2005). The biogas composition and volatile fatty acids concentrations were determined with a gas chromatograph GC 8000Top equipped with a heat conductivity detector HWD 800. The elemental composition of the sludge was assessed by X-ray fluorescence analysis, using the ARL 9400 XP sequential WD-XRF spectrometer. It is equipped with an Rh anode end-window X-ray tube type 4GN fitted with a 75 μm Be window. All peak intensity data were collected in vacuum by WinXRF software.

Foaming potential (FP) and foam stability

These parameters were assessed as additional characteristics of the digested sludge because of the increased presence of filamentous bacteria that cause sludge foaming. It has been reported that anaerobically treated excess activated sludge exhibits a tendency to cause digester foaming. In order to describe and compare foam quantity and quality a so called 'bubble test' has been developed (Zaplatilkova 2004). The testing was based on the test procedure described by Pagilla *et al.* (1996) and modified with respect to the character of the anaerobic sludge. The final test was carried out by bubbling 1 litre of sludge using nitrogen with the flow rate of 1 L/min in 2 L volumetric cylinders. The level of foamy sludge was recorded after 5 minutes of bubbling and the FP was calculated from this value (Equation (1)). Then, 5 min after the gas flow was stopped the level of foamy sludge was recorded again to calculate the index of stability (IS), (Equation (2)). FP describes the capacity of sludge to create foam. IS provides information about the stability of the foam created.

$$FP = \frac{V_5}{V_0} \tag{1}$$

$$IS = \frac{(V_{ST} - V_0)}{(V_5 - V_0)} * 100 \tag{2}$$

where V_0 – volume of sludge at the beginning of measuring (usually 1 L); V_5 – volume of foamy sludge after 5 min of bubbling; V_{ST} – volume of foamy sludge 5 min after the gas flow is stopped.

Microscopy

Microscopic analysis of the sludge samples was performed in accordance with Jenkins *et al.* (1993). The microscopic examination in wet mounts (Olympus BH2-RFCA – ×125, ×250 magnification and phase contrast ×1250 magnification) was aimed at examining basic biomass characteristics

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(morphological properties, structure, presence and types of protozoa/metazoa and zoogloeal colonies). Gram and Neisser staining procedures were applied to determine the abundance of filamentous microorganisms and to identify them.

CST

Capillary suction time (CST) is a simple parameter and a method used for the characterization of sludge dewaterability. The CST is the time necessary to collect a unit volume of filtrate of the sludge undergoing filtration in a standard-sized CST funnel, when placed upon a standard grade of chromatography paper. The CST test was found to be accurate, if the product of solid concentration and specific resistance to filtration is of interest (Scholz 2005). The original circular set-up introduced by Baskerville & Gale (1968) and the Whatman-17 filter paper were used to conduct the measurement.

Table 1 | Comparison of digestion results in final period of experiment (average value and standard deviation)

Parameter	Microaerobic digester R1	Anaerobic digester R2
Volumetric loading rate – COD (g/L.d)	2.0	2.0
Biogas production (L/d)	4.62 ± 0.46	3.67 ± 0.52
Air dose (L/d)	1.6	–
CH_4 production (L/d)	2.33 ± 0.24	2.41 ± 0.35
Specific CH_4 production (L/kg VSS_{added})	188 ± 21	195 ± 28
VSS degradation efficiency (%)	63.6 ± 4.1	65.2 ± 4.7
H_2S removal efficiency (%)	99.7 ± 0.2	–
ORP_H (mV)	– 507 ± 7	– 547 ± 5

RESULTS AND DISCUSSION

Digestion performance

When comparing the performance of both microaerobic and anaerobic digesters it can be stated that, despite the air dosing, the production of methane is comparable in both digesters. Biogas production is higher in the microaerobic digester because the rest of the air supplied – i.e. nitrogen – remains in the biogas. However, methane production is similar in both digesters. Furthermore, the sludge digestion efficiency expressed by VSS degradation efficiency is not significantly different as shown in Table 1. Apparently lower VSS degradation could be a consequence of the potentially higher growth and biomass yield of facultative bacteria. A fundamental difference is in the hydrogen sulfide content in biogas; thanks to microaeration more than 99% of H_2S was removed from the biogas in the final period of digester operation. This confirms the desulfurization potential of the microaerobic technology as previously reported (Jenicek *et al.* 2008; Fdz-Polanco *et al.* 2009) being also effective for the specific composition of treated sludge such as high Fe and S content. Significant differences were confirmed (ORP_H value) in the ORP corresponding to the report by Khanal & Huang (2003). The comparison of H_2S concentrations in biogas during all experimental periods is shown in Figure 1.

Biogas composition

As mentioned earlier the H_2S concentration was the main difference in biogas quality. Even extremely high hydrogen sulfide concentrations were almost completely removed. Another difference was caused by the remaining nitrogen

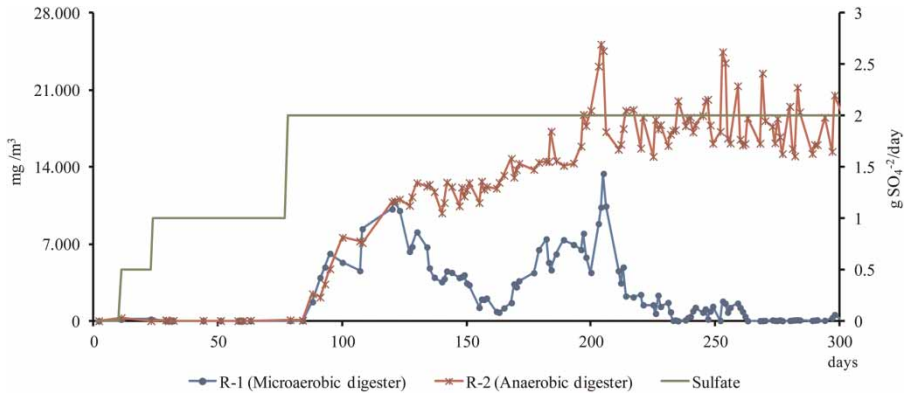


Figure 1 | Hydrogen sulfide concentration in biogas and additional sulfate dose in both digesters.

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gas from the supplied air, which is inert and dilutes all the other components of biogas, especially methane, whose concentration decreased from 63% to 50%. If the decrease of methane concentration is not acceptable, then the dilution can be avoided by substituting the air with oxygen. A slightly lower ratio of CH_4/CO_2 in biogas from the microaerobic digester indicates that a small portion of oxygen is consumed during the oxidation of organic matter. The content of nitrogen was surprisingly high in anaerobic digester. Here, the nitrogen could originate from air getting into the digester with the treated sludge doses, from denitrification and/or from the decomposition of nitrogenous organic compounds. The average biogas composition in the final experimental period is illustrated in Table 2.

Table 2 | Comparison of biogas quality in final period of experiment (average value and standard deviation)

Parameter	Microaerobic digester R1	Anaerobic digester R2
CH_4 (vol.%)	50.4 ± 1.3	63.0 ± 1.2
CO_2 (vol.%)	24.8 ± 2.1	29.0 ± 0.9
N_2 (vol.%)	25.0 ± 2.2	4.8 ± 0.2
H_2 (vol.%)	< 0.1	< 0.1
H_2S (mg/m^3)	51.7 ± 14.4	17,600 ± 490

Sludge liquor composition

The sludge liquor composition differed significantly in many parameters as shown in Table 3 and Figure 2, the most important difference being that of soluble COD during the whole duration of the experiment – it was about 3 g/L lower in the microaerobic digester. The reason for such a result may lie in better degradability of some organic compounds when anaerobic and aerobic conditions are combined. Volatile fatty acids (VFA) concentration was found to be 3.5x higher in the anaerobic digester compared to the VFA concentration in the microaerobic digester. However, that only causes the COD difference in the range of tens of milligrams. Earlier, it had already been found to be true for adsorbable organic halogens (AOX) compounds for example. (Jenicek *et al.* 2008). Abnormally high concentration levels of N_{ammon} (sum N-NH_4^+ and N-NH_3) appeared in both reactors because highly thickened waste activated sludge was digested. However, the N_{ammon} concentration in the microaerobic digester was always slightly lower, about 4% on average. Microaerobic conditions also brought about an expected effect – a lower soluble sulfide concentration, which is in direct relation to highly efficient biogas desulfurization. On the other hand, thiosulfate and sulfate concentrations were higher in R1, but both concentrations can be classified as negligible in comparison with the total sulfur content in the digesters. The microaerobic digester had its lower maximum of the volatile fatty acid concentration when compared to the anaerobic digester during the period in which the digestion process was negatively affected in both digesters, 100 versus 200 mg/L.

Table 3 | Sludge liquor quality in final experimental stage (average value and standard deviation)

Parameter	Microaerobic digester R1	Anaerobic digester R2
pH	8.5 ± 0.1	8.6 ± 0.1
Soluble inorganic salts (g/L)	8.7 ± 0.6	8.8 ± 0.5
Soluble COD (g/L)	5.8 ± 0.4	8.7 ± 0.5
Volatile fatty acids (mg/L)	16.6 ± 0.6	58.8 ± 1.9
Soluble sulfide (mg/L)	162 ± 31	506 ± 50
Thiosulfate (mg/L)	3.5 ± 0.9	1.9 ± 0.2
Sulfate (mg/L)	19 ± 2	37 ± 2
N_{ammon} (g/L)	2.88 ± 0.10	3.00 ± 0.10
P-PO_4^{3-} (mg/L)	55 ± 9	65 ± 10

Sludge composition

No significant differences were found in sludge composition except for sulfide and sulfur content as shown in Tables 4 and 5. The sulfide content is lower in the microaerobic sludge due to its oxidation to elemental sulfur and for the same reason the elemental analysis of the sludge found the higher total sulfur content in the microaerobic sludge. Indications of the slightly higher VSS content in biomass of the microaerobic digester could be a consequence of the potentially higher growth and biomass yield of the facultative bacteria as mentioned in the comment on the performance of the digesters.

Small differences were found in the elemental composition of the digested sludge from microaerobic and anaerobic reactors (Table 5). The expected increase of sulfur content due to the precipitation of elemental sulfur in the microaerobic sludge was confirmed. However, this increase was not particularly high because the sludge containing the accumulated sulfur was continuously removed

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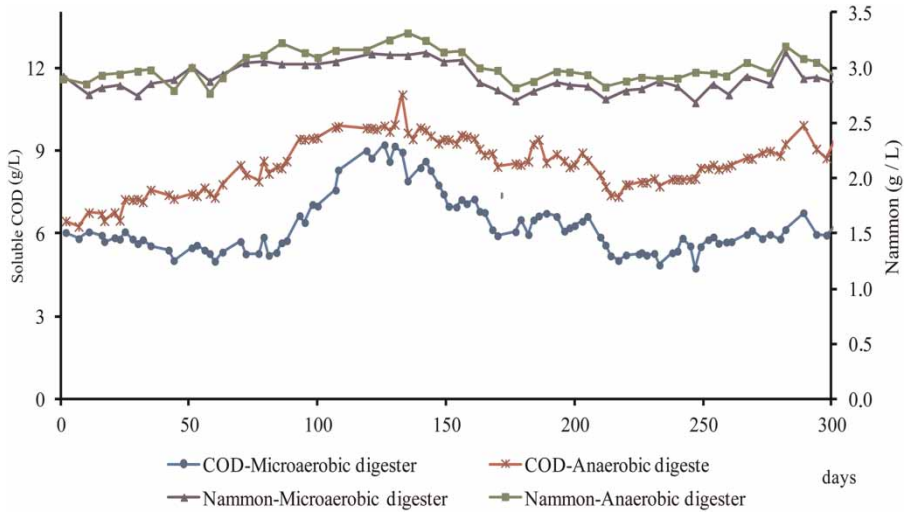


Figure 2 | Differences in soluble COD and N_{ammon} concentrations in both digesters.

Table 4 | Digested sludge composition in final stage of experiment (average value and standard deviation)

Parameter	Microaerobic digester R1	Anaerobic digester R2
TSS (g/L)	31.98 ± 1.30	31.66 ± 1.11
VSS (g/L)	17.96 ± 1.02	17.17 ± 1.24
VSS/TSS (%)	50.5 ± 0.8	49.7 ± 1.6
COD (g/L)	37.6 ± 1.9	37.9 ± 1.9
Total sulfide (mg/L)	824 ± 5	1,129 ± 32

Table 5 | Relative elemental composition of digested sludge from microaerobic and anaerobic reactors (major elements of mineral fraction) 300th day of operation

Element	Microaerobic sludge (%)	Anaerobic sludge (%)	Ratio anaerobic/ aerobic
Fe	20.68	20.48	1.01
P	16.49	16.6	0.99
Ca	11.37	11.62	0.98
S	10.49	8.85	0.84
Na	16.87	15.78	1.07
Al	9.47	9.38	1.01
Si	7.34	7.41	0.99

from the digester. Even at extremely high H_2S concentrations, the maximum daily production of elemental sulfur was 35 mg. High Fe content is caused by the Fe^{3+} addition in the wastewater treatment plant, where the treated activated sludge originates from. The presence of iron also contributes to H_2S removal in both reactors due to the precipitation of FeS.

Foaming

During several test periods, the treated waste activated sludge contained increased levels of filamentous bacteria and so the risk of digester foaming was greater. Initially, when launching microaerobic digester operation, there was concern about the possible stimulation of aerobic bacteria from the activated sludge. The presence of oxygen might lead to the survival of aerobic filamentous bacteria in the microaerobic digester and consequently to the stimulation of foam production. Eventually, our concern about the possible undesirable behaviour of aerobic filamentous bacteria in microaerobic conditions was proven by the results to be unsubstantiated.

The FP of the microaerobic digester was never higher than that of the reference digester during the whole experiment. The difference between the microaerobic and anaerobic digesters was distinct, especially during the period with a high FP. The maximum FP was 3.2 and 2.6 for the anaerobic and microaerobic reactor, respectively. In addition, the foam rising in the anaerobic reactor was much more stable in comparison to the foam originating from the microaerobic reactor. The index of stability IS was 96% in the anaerobic digester and 53% in the microaerobic digester. Better degradation of the activated sludge filaments (with dominance of *Microthrix parvicella*) during the microaerobic digestion can be illustrated by microscopic images of the digested

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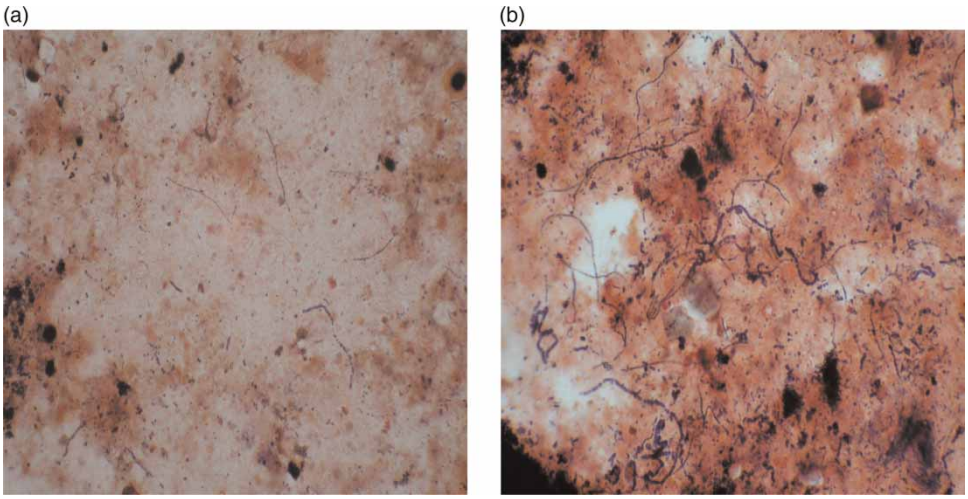


Figure 3 | Microscopic pictures of digested sludge a – microaerobic and b – anaerobic, showing different abundance and morphology of filaments (Gram staining, direct light, enlargement 1250x magnification).

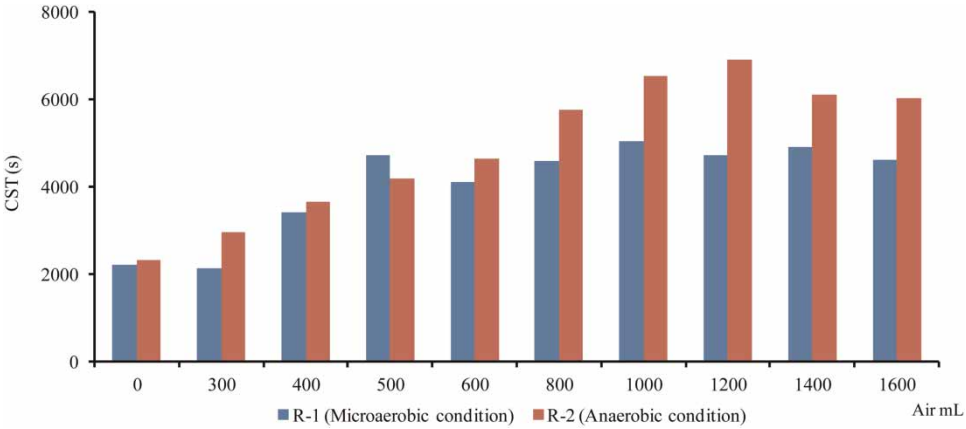


Figure 4 | Comparison of capillary suction times (CST) for the sludge of both digesters.

sludge in Figure 3 with filament fragments being longer, more frequent and more compact in the anaerobic sludge.

Evaluation of dewaterability by CST

Dewaterability was characterized by the periodic CST measurement of sludge samples from both digesters. The results are shown in Figure 4 and indicate that dewaterability of the microaerobic sludge is slightly better especially at higher air doses. Relations between this result and higher concentration of soluble COD in R2 should be studied and explained in further studies. The CST values are relatively high due to the specific character of the treated sludge and due to high TSS and soluble salts concentrations of 30–33 g/L and 8.0–9.5 g/L respectively.

CONCLUSIONS

The following conclusions can be drawn based on the presented results of sludge quality monitoring in anaerobic and microaerobic digesters:

- Sludge quality in both reactors is similar, only small differences were found in sludge composition.
- Slightly higher content of the total S and lower content of the S-S^{2-} in the microaerobic sludge indicates the accumulation of elemental S.

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- The composition of the sludge liquor differs mainly in the soluble COD concentration, which is significantly lower, by 33% on average, in microaerobic digestion.
- The sludge digested under the microaerobic conditions manifested lower FP and foam stability. The dewaterability characterized by CST was also better for the microaerobic sludge.

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